DemandIQ: AI-Driven Retail Demand Forecasting and Inventory Optimization System

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Abstract

Small retail stores often struggle to maintain optimal inventory levels, leading to stock shortages or excess unsold goods. This project, DemandIQ, presents an AI-driven retail demand forecasting and inventory optimization system that predicts future product demand using historical sales data. Existing inventory management systems mainly rely on manual estimation and traditional forecasting methods, which often lead to inaccurate demand predictions, stock shortages, overstocking, increased inventory costs, and poor business decision-making. By combining machine learning and time-series forecasting techniques, the system analyzes past sales patterns to improve demand prediction accuracy and provide optimal restocking recommendations. This helps retailers reduce stockouts, minimize overstocking, and manage inventory more efficiently. The proposed system demonstrates how artificial intelligence can be applied to real-world retail challenges, providing a practical, low-cost solution for small businesses to optimize inventory management and support better business decision-making.

Keywords: Inventory Management, Demand Forecasting, Machine Learning, Time-Series Forecasting, Retail Analytics, Predictive Modelling.

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ISBN No.: 978-93-344-7370-4